

FLAWFIX SECUREASSIST

Comprehensive Security Vulnerability Audit, Explainable AI & Patch Verification Report

Target File: pytest.py	Language: PYTHON
Project: FlawFix Workspace	Audit Date: 2026-10-07 11:23:53
Report ID: RPT-408111B2	Analysis Engine: FlawFix LLVM IR + Gemini XAI Dual-Engine

1. Executive Security Assessment

The provided Python code correctly utilizes parameterized queries for SQLite interactions. By passing the user input as a tuple to the execute method, the sqlite3 library handles the sanitization and escaping of the input, effectively preventing SQL injection attacks. The code follows secure coding practices for database interactions.

TOTAL FLAWS	CRITICAL	HIGH	MEDIUM	LOW	VERIFIED PATCHES
0	0	0	0	0	1

2. Detected Vulnerability Inventory

✓ No security vulnerabilities detected. Code successfully passed semantic audit.
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4. Automated Patch Verification Audit

The following patches have been compiled and verified through the FlawFix dual-engine pipeline:

Vulnerability ID	Verification Gate Status	Compiler Audit Details
VULN-001	✓ VERIFIED & VALIDATED	The original code was vulnerable to SQL injection due to direct string concatenation of user input into the SQL query. The patch correctly implements parameterized queries using the '?' placeholder, which forces the sqlite3 library to treat the input as data rather than executable SQL code, effectively neutralizing the injection vector.

5. AI Code Optimization Recommendations

Optimization Focus	Description & Efficiency Advantage	Expected Benefit
Use Context Managers	Wrap the database connection in a 'with' statement to ensure the connection is automatically closed even if an exception occurs.	Improves resource management and prevents potential connection leaks.

Input Sanitization	While parameterized queries prevent SQL injection, adding basic length validation or regex filtering on the 'username' input can prevent denial-of-service attacks involving excessively large strings.	Increases application robustness against malformed input.
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